7.2 → Evaluating reserving techniques

7.2.1 → Evaluating reserving techniques

→ Monitoring unpaid claims estimates

→ After making the final unpaid claims estimate, it is crucial to monitor its accuracy. If actual claim development deviates significantly from expected development, it may be beneficial to re-evaluate the estimate. Comparing actual to expected paid or reported claims can also be a useful diagnostic tool.

→ We can compute the expected emergence, the total expected reported claims reported between two different maturities, using the formula below:

$$\text{Expected Rept Claims between } t_1 + t_2 = \text{Estimated IBNR} \times \%\text{ Expected to emerge}$$

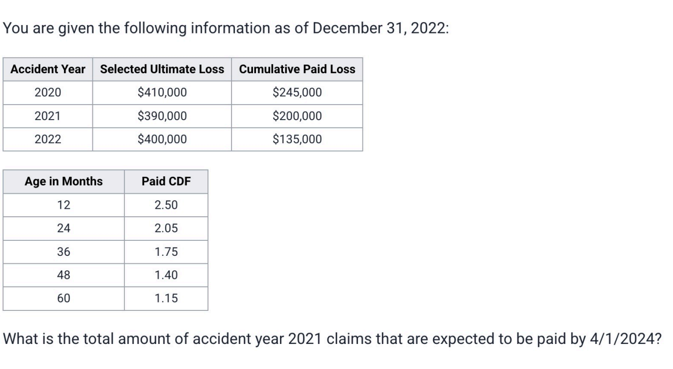
$$= \left[\text{Estimated ult claims} - \text{Latest Rept claims}\right] \times \left[\frac{\%\text{ Rept}_{t_2} - \%\text{ Rept}_{t_1}}{1 - \text{Latest }\%\text{ Rept}}\right]$$

$$= \downarrow \quad \downarrow \quad \times \left[\frac{\frac{1}{\text{rept CDF}_{t_2}} - \frac{1}{\text{rept CDF}_{t_1}}}{1 - \frac{1}{\text{Latest rept CDF}}}\right]$$

< then paid (...) for paid method >

%/factor corresponds to the age of the respective AY relative to the valuation date

→ Example →



You are given the following information as of December 31, 2022:

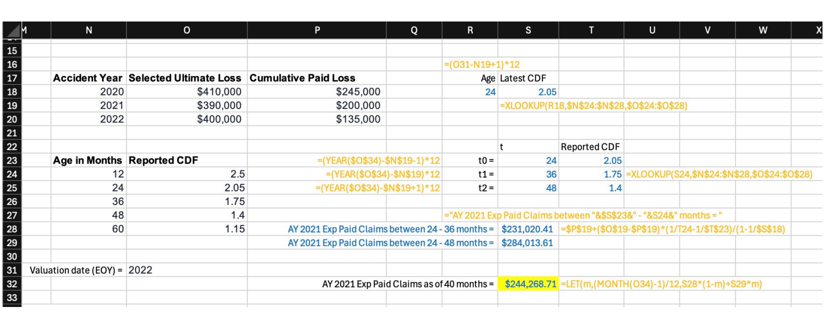
Accident Year	Selected Ultimate Loss	Cumulative Paid Loss
2020	$410,000	$245,000
2021	$390,000	$200,000
2022	$400,000	$135,000

Age in Months	Paid CDF
12	2.50
24	2.05
36	1.75
48	1.40
60	1.15

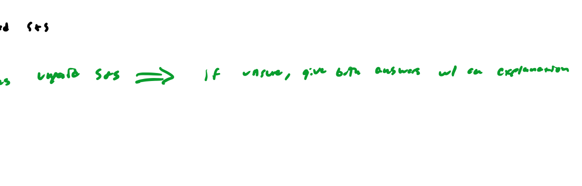
What is the total amount of accident year 2021 claims that are expected to be paid by 4/1/2024?

* think I can logic my way through these ones

→



	M	N	O	P	Q	R	S	T	U
15									
16						=(O31-N19+1)*12			
17		Accident Year	Selected Ultimate Loss	Cumulative Paid Loss		Age	Latest CDF		
18		2020	$410,000	$245,000		24	2.05		
19		2021	$390,000	$200,000			=XLOOKUP(R18,$N$24:$N$28,$O$24:$O$28)		
20		2022	$400,000	$135,000					
21									
22							t	Reported CDF	
23		Age in Months	Reported CDF	=(YEAR($O$34)-$N$19-1)*12		t0 =	24	2.05	
24		12	2.5	=(YEAR($O$34)-$N$19)*12		t1 =	36	1.75	=XLOOKUP(S24,$N$24:$N$28,$O$24:$O$28)
25		24	2.05	=(YEAR($O$34)-$N$19+1)*12		t2 =	48	1.4	
26		36	1.75						
27		48	1.4			="AY 2021 Exp Paid Claims between "&$S$23&" - "&S24&" months = "			
28		60	1.15	AY 2021 Exp Paid Claims between 24 - 36 months =			$231,020.41	=$P$19+($O$19-$P$19)*(1/T24-1/$T$23)/(1-1/$S$18)	
29				AY 2021 Exp Paid Claims between 24 - 48 months =			$284,013.61		
30									
31		Valuation date (EOY) =	2022						
32				AY 2021 Exp Paid Claims as of 40 months =			$244,268.71	=LET(m,(MONTH(O34)-1)/12,S28*(1-m)+S29*m)	
33									



8.1 → Recoveries: Salvage + Subrogation + reinsurance

8.1.1 → Salvage + Subrogation

→ Final formula

unpaid S+S = Ultimate S+S − Received S+S

or using reported S+S:

unreported S+S = Ultimate S+S − Reported S+S

"recoverable" → recoverable S+S means unpaid S+S ⟹ if unsure, give both answers w/ an explanation

→ Methods

→ #1) Development method

⟹ Perform like usual directly on S+S dollars

→ #2) Ratio method

⟹ Step 1 → Create S+S ratio triangle

Step 2 → Develop the gross paid claims triangle

Step 3 → Estimate unpaid S+S

→ Summary

→ The ratio method offers two distinct advantages over the development technique:

→ Less leverage in development factors. The ratio approach offers the advantage of generating development factors that are less heavily influenced by reclaimed S+S dollars.

→ Refined selection of S+S ratios. Another advantage is linked to the process of selecting ultimate S+S ratios for recent years. By considering the average of ultimate S+S ratios from previous years, excluding the current year, a more representative + reliable ultimate S+S ratio can be chosen, leading to improved accuracy in estimating future claims.

→ One disadvantage of the ratio method is that an error in selecting ultimate losses will likely lead to an error in the S+S estimates as well.

→ S+S ratio method →
1) calculate S+S ratios
2) calculate LDFs
3) select LDFs + calc CDFs
4) develop S+S ratios → select ult ratio (multiple options, just justify)
5) use (4) + × by rept losses gross of S+S
6) manipulate to get final answer of interest

advantages
1) can use actuarial judgement to select ult S+S ratio
2) more stable than dev method
⟹ i.e.) LDFs using this approach are less leveraged than LDFs of S+S $

amount recoverable = est ult S+S − already received

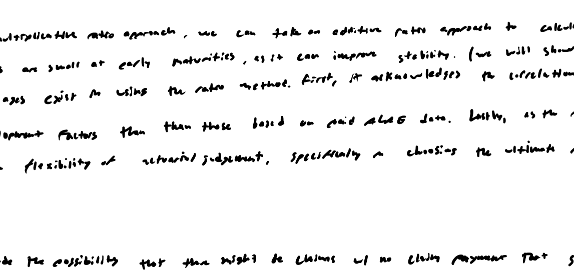
8.1.2 → Reinsurance

→ Two types → quota share, e.g.) 70%/30% ↳ we pay, they pay

→ excess of loss → there is a cap of some kind, depending on how it is defined

→ look at net-to-gross triangles to determine which type of treaty is in place

→ DRAW out reinsurance contracts ⟹ Do one-at-a-time + in pieces ↳ paid + unpaid (IBNR)



8.2 → Estimating unpaid allocated claims adjustment expenses

8.2.1 → Estimating unpaid ALAE

→ Estimating unpaid ALAE

→ The two methods discussed in section 8.1, the development technique + the ratio technique, are also applicable to ALAE estimation. Paid ALAE can be estimated using the development technique; if case outstanding data is available, then reported ALAE can also be estimated using the development technique.

→ In addition to a multiplicative ratio approach, we can take an additive ratio approach to calculate ALAE. This is called additive development + is best used when ratios are small at early maturities, as it can improve stability. (we will show how to use the additive approach in the example below). Several advantages exist in using the ratio method. First, it acknowledges the correlation between ALAE + claims. Second, it results in less leveraged development factors than those based on paid ALAE data. Lastly, as the ratios can serve as a diagnostic tool, the ratio method allows for the flexibility of actuarial judgement, specifically in choosing the ultimate ALAE ratios from the latest years w/in the experience period.

→ Disadvantages include the possibility that there might be claims w/ no claim payment that still have a large ALAE. Another disadvantage is that an error in estimating ultimate claims can lead to an error in the ultimate ALAE estimate.

8.3.1 → Dollar-based techniques

→ Overview → All dollar-based techniques assume that ULAE costs closely follow claims costs, where claim costs include losses + ALAE, but exclude ULAE. Specifically, we assume that ULAE expenditures follow the general timing of claims (reported or paid), + are proportional to their size, e.g. a claim of $1000 requires 10 times more ULAE than a claim of $100.

→ The four dollar-based techniques we will discuss are:
- → the classical (traditional) approach
- → the Mango-Allen refinement
- → the Kittel refinement
- → the generalized approach

→ Notes → Dollar-based techniques all relate ULAE dollars to claims dollars, hence the term "dollar-based".

→ Classical technique

→ In the classical technique, also called the traditional technique, unpaid ULAE is estimated by first creating a ratio of calendar year paid ULAE to calendar year paid claims:

$$W_c = \frac{\text{CY paid ULAE}}{\text{CY paid claims}}$$

* → unpaid ULAE = $W_c \times [\text{pure IBNR} + 0.5(\text{case outstanding} + \text{IBNER})]$

→ In general, to estimate unpaid ULAE under the classical technique, follow these four steps:

→ 1) Calculate historical ratios. Calculate the historical ratio of paid ULAE-to-paid claims, W_c, for each calendar year.

→ 2) Analyze ratios. Analyze the historical ratios for any trends or patterns.

→ 3) Select a ULAE-to-claims ratio. Choose a ULAE-to-claims ratio, W_c^*.

→ 4) Apply the selected ratio. Use the equation above to estimate the unpaid ULAE.

→ Key assumptions of the classical technique

→ As we have seen, the classical technique makes three key assumptions:

→ The ratio of paid ULAE-to-paid claims is stable enough such that it can reasonably estimate the ultimate ULAE-to-ultimate claims relationship.

→ The future activity + cost associated w/ managing claims that have not been reported yet are directly proportional to pure IBNR, + those that have been reported but not yet closed are directly proportional to case outstanding + IBNER.

→ 50% of ULAE costs occur when a claim is opened + 50% when a claim is closed.

→ When the classical technique works + when it doesn't

→ The classical technique provides good results for very short-tailed, stable LOBs. It doesn't work well in:

- → long-tail lines
- → Periods marked by shifting inflationary pressures, whether historical or anticipated
- → Instances when an insurance company undergoes a shift in volume by either expanding or contracting rapidly
- → Cases where the 50/50 assumption does not accurately reflect the claims process

→ Mango-Allen refinement

→ The Mango-Allen refinement is nearly identical to the classical technique w/ one exception: instead of using the actual calendar period claims, expected paid claims are used.

$$W_M = \frac{\text{CY paid ULAE}}{\text{CY expected paid claims}}$$

→ Kittel refinement

→ Consider an insurer that is rapidly growing. During this period, ULAE costs will increase, yet it is reasonable to expect that a majority of claims are, most likely, payable in the future. As a result, under the classical technique, the true ULAE-to-claims relationship may be distorted. This is b/c the classical technique charges ULAE payments when payments are made + does not take into account the reality that ULAE is incurred regardless if payments are made or not.

→ The Kittel refinement is an approach that attempts to deal w/ this discrepancy.

→ Key assumptions of the Kittel refinement

→ In the Kittel refinement, the following two key assumptions differ from the classical technique:

→ ULAE is sustained as claims are reported even if no claim payments are made.

→ ULAE payments for a specific CY are related to both the reporting + payment of claims

→ Aside from this, the Kittel refinement assumes that 50% of ULAE is paid when opening a claim + 50% is paid when closing a claim, similar to the classical technique. Thus, using the Kittel approach:

- → For claims which are opened + closed during the year, 100% of ULAE is paid
- → For claims that are already open + closed during the year, 50% of ULAE is paid
- → For claims that are opened during the year + remain open, 50% of ULAE is paid

→ Thus, in this approach, the ULAE-to-claims ratio is instead defined as:

$$W_K = \frac{\text{CY paid ULAE}}{0.5(\text{CY paid claims} + \text{CY incurred claims})}$$

→ When the Kittel refinement works + when it doesn't

→ The Kittel refinement tackles the distortion caused by the classical technique for a growing company. However, it still sticks to the conventional 50/50 assumption for ULAE expenditure. In other words, it does not account for the specific distribution of ULAE costs across opening, maintaining, + closing claims, which can differ among insurers. Additionally, the Kittel refinement does not adequately handle the case when inflation rates vary between ULAE + the [illegible]

→ Generalized approach

→ The generalized Kittel approach, or just generalized approach, is a generalization of the classical + Kittel refinement approaches. Developed by Conger + Nolibos, one of its key features is that it assigns different weights to claims depending on their stage of development.

→ Specifically, in the general approach, three weights, $U_1, U_2, + U_3$ are defined as follows:

- → U_1 is the percent of ultimate ULAE spent in opening claims
- → U_2 — — — — — — — — — — maintaining claims
- → U_3 — — — — — — — — — closing claims
- → $U_1 + U_2 + U_3 = 100\%$

→ Similar to previous approaches, we estimate the ULAE-to-claims ratio, W, for each CY as the paid ULAE, M, divided by the claims basis, B.

$$W = \frac{M}{B}$$

where B is calculated as:

$$B = (U_1 \times R) + (U_2 \times P) + (U_3 \times C)$$

+ R, P, + C are defined as:

- → R = ultimate cost of claims reported during the CY
- → P = paid claims during the year
- → C: ultimate cost of claims closed during the CY

→ After calculating W for several CYs, we decide upon a suitable W^*. W/ this value selected, the refined method by Conger + Nolibos to estimate the unpaid ULAE is analogous to the Bornhuetter-Ferguson approach:

$$\text{unpaid ULAE} = W^* \times (L - B)$$

where L is the ultimate losses for an AY.

→ The second approach Conger + Nolibos develop is analogous to the expected claims approach: unpaid ULAE = $(W^* \times L) - M$

→ Conger + Nolibos do not favor this method b/c it is difficult to calculate ULAE on an AY basis, which is necessary for alignment w/ other analyses. The method also does not adapt well to changes in the ULAE-to-claims ratio over time.

→ A third method they describe is similar to the development approach: unpaid ULAE = $M \times \left(\frac{L}{B} - 1\right)$

→ Conger + Nolibos point out that this approach is faulty for two reasons: it can be overly sensitive to random fluctuations in ULAE emergence, in addition to the practical difficulty of tracking ULAE payments for specific time frames.

→ Key assumptions of the generalized approach

→ The generalized approach makes several key assumptions:

- → ULAE expenses are directly tied to the dollar amount of claims being handled.
- → ULAE amounts spent when opening claims are proportional to the ultimate cost of those claims when reported.
- → ULAE amounts spent during the maintenance of claims correspond to the payments made during their handling.
- → ULAE expenses for closing claims are proportional to the final cost of the claims being closed

→ generalized approach for estimating ULAE

CY	M Paid ULAE	R Est ult Reported loss	P Paid loss + ALAE	C Est ult loss on closed claims	B Sumproduct (R P C, U1 U2 U3)	W M/B

U_1 = Open % U_2 = Maintain % U_3 = Close %

select W^*

B-F method

⟹ unpaid ULAE = $W^* (L - B)$
↓ total est ult loss ↳ ΣB

→ Issue w/ dollar-based techniques for ULAE
⟹ ignores claim counts: in reality more claims tend to lead to higher ULAE

→ Simplified generalized approach

→ B/c it is difficult to estimate R + C, the simplified generalized approach was developed as an alternative method. This approach estimates B, which is denoted as $B_{estimate}$, by replacing R + C + letting $U_3 = 0$:

$$B_{estimate} = (U_1 \times L) + (U_2 \times P)$$

Then W is defined as:

$$W = \frac{M}{B_{estimate}}$$

→ A key result is that, when using the Bornhuetter-Ferguson method applied to the generalized approach, the estimated unpaid ULAE under this simplified approach is:

$$\text{Unpaid ULAE} = W^* \times [U_1 \times \text{Pure IBNR} + U_2 \times (\text{case outstanding} + \text{IBNER} + \text{pure IBNR})]$$

$$\downarrow \quad = W^* \times [\text{Pure IBNR} + U_2 \times (\text{case outstanding} + \text{IBNER})]$$

→ Note that when $U_1 = U_2 = 0.50$ we get the exact same formula as used in the classical + Kittel approaches. In other words, the classical + Kittel approaches are just special cases of the generalized approach.

8.3.2 → Count-Based techniques

→ Overview → Dollar-based techniques make several simplifying assumptions. In reality, processing 100 claims of size $1000 will cost more in ULAE than one claim of size $100,000. In addition, sudden drops (or increases) in claim numbers or values are unlikely to lead to an immediate reduction (or increase) in overhead expenses or claims management personnel. Dollar-based techniques do not account for both of these ideas.

→ Count-based techniques, on the other hand, assume that the cost of each claims transaction is the same regardless of claim size, + also assume that a claim which is open longer will cost more. There are six different approaches that we will discuss:

→ Generalized approach

→ The generalized approach can be applied to claim counts as well. Again, letting M be the amount of ULAE payments, we can calculate the ULAE-to-claim counts ratio, w, as:

$$w = \frac{M}{b}$$

where the claim count basis, b, can be calculated as:

$$b = (v_1 \times r) + (v_2 \times o) + (v_3 \times c)$$

which relies on the following inputs:

- → r = reported claim counts
- → o = open — — —
- → c = closed — — —
- → v_1 = relative cost of handling the reporting of a claim
- → v_2 = — — — — managing an open claim
- → v_3 = — — — — closing a claim

→ Unlike the dollar-based generalized approach, the sum of $v_1 + v_2 + v_3$ does not have to equal 100%.

→ Next, as we did before, select w^*, or a series of w_i^* terms, which can be different for each CY to take inflation into account. Finally, the unpaid ULAE can be estimated as:

$$\text{Unpaid ULAE} = \sum_i w_i^* \times [(v_1 \times r_i) + (v_2 \times o_i) + (v_3 \times c_i)]$$

where:

- → r_i is the number of claims reported in CY i
- → o_i — — — — — open claims at the end of CY i
- → c_i — — — — — claims to be closed during CY i

→ Misc

→ Think about actual things done here at work when answering written questions

→ If confused on a worded question, break it down + understand the pieces, so can use logic to hopefully get somewhere
↳ + other concepts

→ STATE ASSUMPTIONS, e.g. I am assuming the question is asking for the excess loading factor (ELF) for losses excluding cat losses (xcat)